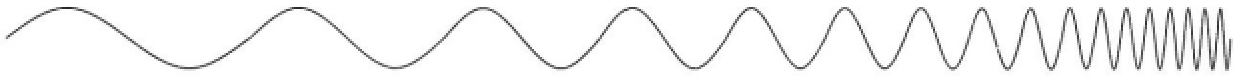
(ANSWER)

Electromagnetic wave



Radio waves (無線電波)	Microwaves (微波)	Infra-red (紅外輻射)	Visible light (可見光)	Ultra-violet (紫外輻射)	X-rays (X射線)	Gamma rays (伽偶射線)
~ 10 m	~ 10^{-2} m	~ 10^{-5} m	10^{-7} m	~ 10^{-8} m	~ 10^{-10} m	~ 10^{-12} m
			red (700 nm) violet (400 nm)			

The longest wavelength

The lowest frequency

all electromagnetic waves have the same speed in air or vacuum

The shortest wavelength

The highest frequency

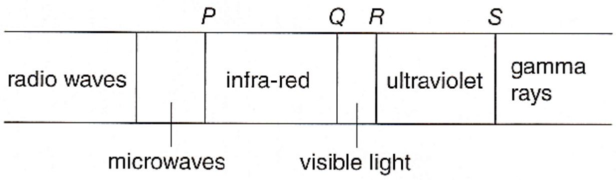
- They are **transverse waves.**
- They **carry energy.**
- They do not require a medium to travel, and **can travel in a vacuum.**
- In a *vacuum*, their **speed** ***c*** is fixed at $\mathbf{3 \times 10^8\ m\ s^{-1}}$. In a material, they travel slower at *various* speeds.
- They obey $\boldsymbol{v = f\lambda}$.
- They show **reflection, refraction, diffraction and interference.**

	Source	**Detector**	**Uses**	**Wavelength**
Radio waves	Radio and TV transmitter	Antenna	● Radio/TV broadcasting Mobile phone communication	$10^{-1} – 10^{4}$ m
Microwaves	Microwave transmitter	Microwave receiver	● Microwave oven ● Satellite communication ● Radar	$10^{-3} – 10^{-1}$ m
Infrared	Objects above **0** K	Skin, phototransistor	● Remote control used in TV ● Auto-focus of camera ● Infra-red thermometer	$10^{-6} – 10^{-3}$ m
Visible light	Hot objects, Sun	Human eye	●	4 x 10^{-7} – 7 x 10^{-7} m
Ultraviolet	UV lamp, Sun	Fluorescent material	● To sterilize drinking water Checking banknotes	$10^{-9} – 10^{-8}$ m
X-ray	X-ray tube	Photographic film	● Production of see-through images (human body or luggage)	10^{-10} m
Gamma ray	Radioactive substance	GM tube, photographic film	● Medical uses ● Sterilization (killing bacteria in food) ● Radiotherapy	10^{-12} m

(ANSWER)

1. Student draws the electromagnetic spectrum. However, the part for X-ray is missing. Where should the part for X-ray be?

A. *P*
B. *Q*
C. *R*
D. *S*



D

2. Which of the following shows the correct order of five electromagnetic waves arranged in the order of decreasing frequency?

A X-rays, infra-red, microwaves, radio waves, ultraviolet
B Ultra-violet, infra-red, radio waves, X-rays, microwaves
C Radio waves, microwaves, ultraviolet, infra-red, X-rays
D X-rays, ultraviolet, infra-red, microwaves, radio waves

D

3. Which of the following statements about electromagnetic waves is/are correct?

(1) The speeds of electromagnetic waves are equal to the speed of light.
(2) All kinds of electromagnetic waves can travel in a vacuum.
(3) Electromagnetic waves are transverse waves.

A (1) only
B (2) only
C (1) and (3) only
D (1), (2) and (3)

D

4. Which of the following statements about electromagnetic waves is **incorrect**?

A They can transmit energy through a vacuum.
B They travel with the same speed in all media.
C They are transverse waves in the electric and magnetic fields.
D They all show reflection, refraction, interference and diffraction.

A *P*
B *Q*
C *R*
D *S*

B

5. A certain medical light has a period T, where 10^{15}T = 1 s, what types of EM wave does it belong to?

A Infra-red
B visible light
C ultra-violet
D X-ray

$$v = f\lambda$$
$$3\times10^{8} = \frac{1}{\frac{1}{10^{15}}}\lambda$$
$$\lambda = 3\times10^{-7}\ \mathrm{m}$$
$\therefore$ It is ultraviolet radiation.

C

(ANSWER)

6. Which of the following is not an application of microwaves?
   A. Radar
   B. Microwave oven
   C. Satellite communication
   D. Thermometer

D

7. Which of the following is not an application of infrared radiation?
   A. Thermometer
   B. Auto-focus camera
   C. Laser pointer
   D. Electric oven

C

8. Which of the following are applications of ultraviolet radiation?
   (1) Sterilization of water
   (2) Remote control
   (3) Checking of banknote
   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3)

B

9. Which of the following type of electromagnetic waves is best used for killing cancer cells?
   A. Gamma rays
   B. Infrared radiation
   C. Visible light
   D. Ultraviolet radiation

A

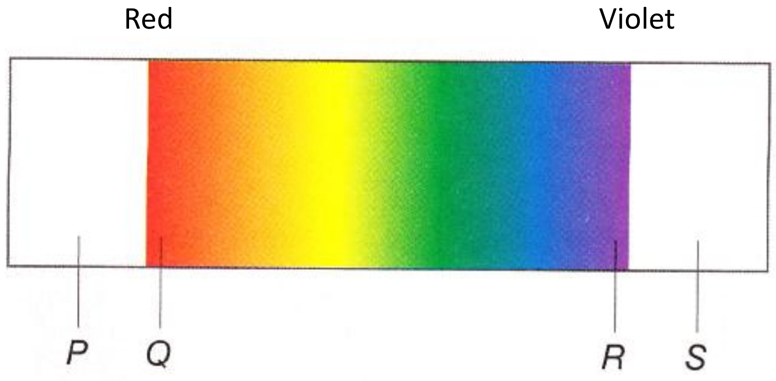
10. Which of the following correctly shows the uses of microwaves, radio waves and ultraviolet radiation?

	Microwaves	**Radio waves**	**Ultraviolet radiation**
A.	Radar	Burglar alarm	Medical diagnosis
B.	Communication	TV broadcast	Checking banknote
C.	Tracer	Sterilizing water	Remote control
D.	Cooking	Radiotherapy	Thermography

B

(ANSWER)

11. Figure below shows part of the electromagnetic spectrum.



(a) Which one has the longest wavelength, P, Q, R or S? P

(b) Which one has the highest speed in a vacuum? Same

(c) What is the speed of Q in a vacuum? $3 \times 10^8\ m\ s^{-1}$

(d) Name P and S.

P: infra-red radiation

S: ultra-violet radiation

12. The electromagnetic spectrum is shown in the figure.

radio waves	*X*	infra-red radiations	visible light	*Y*	X-rays	*Z*

(a) Name the parts *X*, *Y* and *Z*. (3 marks)

(b) (i) Which part of the spectrum has the shortest wavelength? (1 mark)

(ii) Which part of the spectrum has the lowest frequency? (1 mark)

(c) State one application of *Z*. (1 mark)

(a) *X* : microwaves

Y : ultra-violet radiation

Z : gamma rays

(b) (i) Gamma rays

(ii) Radio waves

(c) Any one of the following:

Radiotherapy

Sterilization

Medical imaging

(Or other reasonable answer)

(ANSWER)

13. (a) In Hong Kong, TV signals are broadcasted on UHF (ultra high frequency) at about 500 MHz. Some TV stations in other countries, such as Australia, use signals on VHF (very high frequency) at about 50 MHz.

(Given: speed of electromagnetic wave in air = 3×10^8 m s^{-1})

(i) Name the electromagnetic waves used in TV broadcasting. (1 mark)

(ii) Calculate the wavelengths of VHF and UHF waves at the frequencies given above. (3 marks)

(iii) Will it be easier to receive strong signals on VHF or UHF in an urban area full of tall buildings? Explain your answer. (3 marks)

(iv) For a user 25 km away from a broadcasting station, what is the shortest time the signals take to reach the user? (2 marks)

(b) The table below shows the broadcasting frequencies of some local radio stations.

Radio station	FM	AM
RTHK 1	92.6 – 94.4 MHz	—
RTHK 2	94.8 – 96.9 MHz	—
RTHK 3	—	567 kHz
RTHK 4	97.6 – 98.9 MHz	—
RTHK 5	—	783 kHz
CRHK 1	88.1 – 89.5 MHz	—
CRHK 2	90.3 – 92.1 MHz	—
Metro Radio	99.7 MHz	—
Metro Finance	104 MHz	—
Metro Plus	—	1044 kHz

A radio station uses a wave of wavelength 3.4 m in broadcasting.

(i) Find the broadcasting frequency of the radio station. (1 mark)

(ii) Which radio station is it?

(a) (i) Radio wave

(ii) By $v = f\lambda$,

wavelength of VHF $= \dfrac{v}{f} = \dfrac{3 \times 10^8}{50 \times 10^6} = 6$ m

wavelength of UHF $= \dfrac{v}{f} = \dfrac{3 \times 10^8}{500 \times 10^6} = 0.6$ m

(iii) VHF

VHF waves have longer wavelengths

and diffract more when they encounter buildings.

(iv) Shortest time $= \dfrac{\text{distance}}{\text{wave speed}} = \dfrac{25 \times 10^3}{3 \times 10^8}$

$= 8.33 \times 10^{-5}$ s

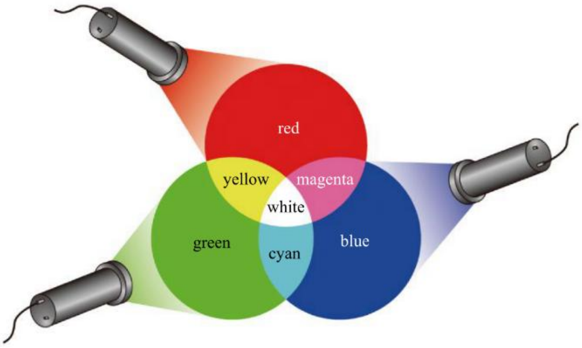
(b) (i) Frequency $= \dfrac{v}{\lambda} = \dfrac{3 \times 10^8}{3.4} = 88.2 \times 10^6$ Hz

(ii) CRHK 1

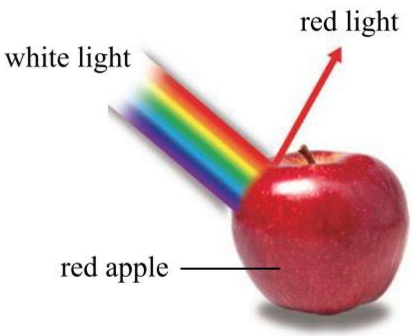
(ANSWER)

- Red, green and blue are called primary colours.

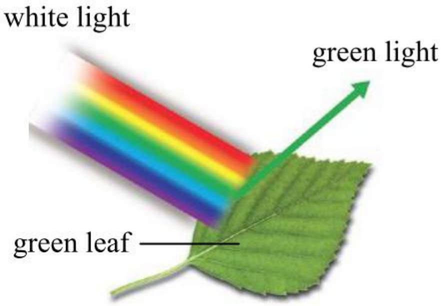
- Red, green and blue lights are called primary coloured lights. We can produce any coloured light by mixing primary coloured lights in different proportions.



- When an object is shone with white light, it absorbs some coloured lights and reflects the rest. The colour of the object is determined by the coloured light it reflects.

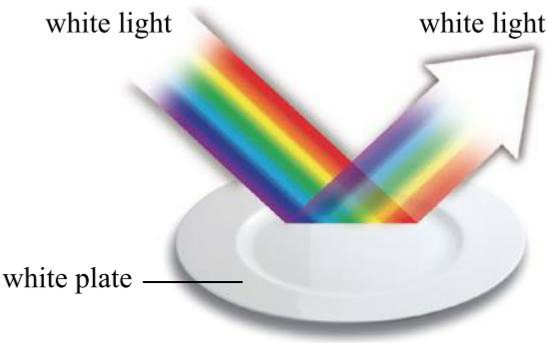


When white light shines upon a red apple, the apple reflects red light and absorbs other coloured lights. This is why it appears red.

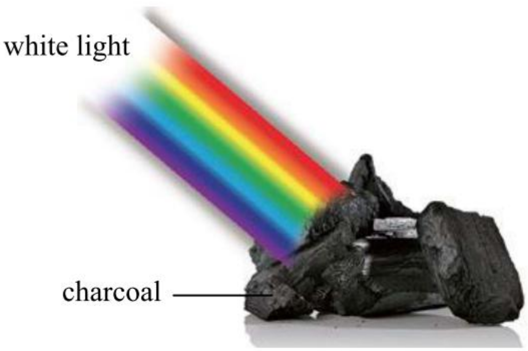


When white light shines upon a green leaf, the leaf reflects green light and absorbs other coloured lights. This is why it appears green.

- Some objects do not absorb any coloured lights but reflect all of them. They appear white when shone with white light. On the other hand, some objects absorb all coloured lights and do not reflect any light. They appear black.



When shone with white light, the plate appears white since it reflects all coloured lights.



The charcoal appears black since it absorbs all coloured lights and does not reflect any light

(ANSWER)

14. Yellow paint absorbs blue light; cyan paint absorbs red light; magenta paint absorbs green light. For this question, assume that white light is composed of red, green and blue lights only.
    (a) Explain why yellow paint appears yellow under white light. (2 marks)
    (b) Explain why magenta paint appears magenta under white light. (2 marks)
    (c) What is the colour of the mixture of yellow and magenta paints? Explain your answer. (4 marks)
    (d) What is the colour of the mix of red, green and blue paints? Explain your answer. (2 marks)

(a) Yellow paint absorbs blue light and reflects red and green lights. (1) The mix of red and green lights gives yellow light. (1)

(b) Magenta paint absorbs green light and reflects red and blue lights. (1) The mix of red and blue lights gives magenta light. (1)

(c) The mixture of yellow and magenta paints absorbs blue (1) and green lights (1). Only red light is reflected. (1) So the mix looks red. (1)

(d) The mix of red, green and blue paints absorbs all red, blue and green lights. (1) So the mix appears black. (1)

